

Algorithmic thinking: selection

Revise 360

Worksheet for the 360° experience: Decision control centre

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Decision control centre. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Apply selection to decisions.
- Handle boundary and equality cases.
- Combine conditions accurately.

Starter: before you put the headset on

A club accepts pupils aged 12 or older. Should age 12 be accepted?

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Key terminology

Selection

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Condition

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The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Two possible paths

Turn right

Key facts in your own words:

Challenge: What happens when available is false?

If available is false, which message is printed?

An IF condition evaluates to...

2 Include the boundary

Turn right

Key facts in your own words:

Challenge: Why does age > 12 not match the rule?

Which condition means score is at least 50?

Which age is excluded by age > 12 but included by age >= 12?

3 Three-way decisions

Turn around

Key facts in your own words:

Challenge: What happens if the draw case is omitted?

A = 7 and B = 7. What is the correct result?

Which result fits a = 9 and b = 4?

4 Both conditions

Turn around

Key facts in your own words:

Challenge: Can this member borrow the unavailable item?

Which permits a loan only if both checks pass?

true AND false is...

5 Either condition

Turn left

Key facts in your own words:

Challenge: Does a person who meets both conditions qualify?

student = false, senior = true. What is student OR senior?

true OR true is...

6 Nested decisions

Turn left

Key facts in your own words:

Challenge: If member is false, is the inner decision reached?

Which describes nesting?

An inner branch inside a false outer IF is...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Match the phrase to its condition.

Match each item to a description from the list below.

$x == 10$ / $x < 10$ / $x >= 10$

Item	Matching description
At least 10	
Below 10	
Exactly 10	

A loan needs a member AND an available item. Member is true and available is false. Result?

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Your score

Copy it from the score screen as soon as it appears.

_____ / 16

Key question

Is age greater than 16 the same as age at least 16?

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Review and improve

Use review mode. Explain one correction and why the correct answer fits.

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Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Write a condition for a temperature from 18 to 24 inclusive. [2]

b) Explain the difference between AND and OR. [2]

c) An algorithm uses $\text{age} > 16$ for a rule of at least 16. Identify and correct the error. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Apply selection to decisions.			
Handle boundary and equality cases.			
Combine conditions accurately.			